MINI PROJECT REPORT

1. Project Objective

The main objectives of this project are:

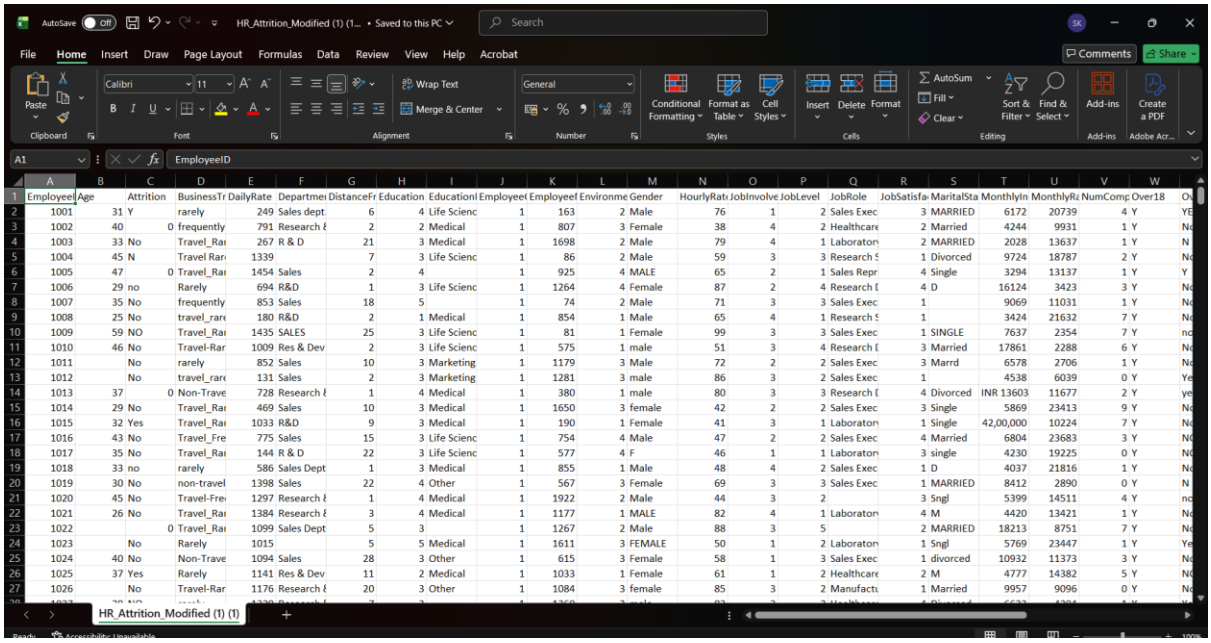
- To perform data preprocessing using Microsoft Excel.
- To clean and transform raw data by removing duplicates and handling missing values.
- To standardize data formats and prepare the dataset for analysis.
- To create meaningful visualizations using Power BI.
- To analyze employee attrition patterns using interactive dashboards.

---

2. Dataset Description

- The dataset used in this project is the HR Employee Attrition dataset.
- Dataset characteristics:

Attribute	Description
Number of Records	**600**
Number of Features	**36**
Data Type	**Numerical and Categorical**
Domain	**Human Resources**



---

3. DataSet:

https://drive.google.com/file/d/1zReA2Et5Kl5pw9CuU7D0j_q4xhmfKRqW/view?usp=drive_link

---

4. Data Pre-processing using Excel

Raw data often contains errors or inconsistencies. Therefore, several preprocessing steps were performed in Microsoft Excel to improve data quality.

4.1 Removing Duplicates

Duplicate records can lead to incorrect analysis results. Excel's Remove Duplicates feature was used to identify and remove repeated records based on employee attributes.

Steps:

- Selected the dataset
- Used Data → Remove Duplicates
- Verified that duplicate rows were removed

4.2 Handling Missing Values

Missing values were identified using Go To Special → Blanks.

Different techniques were applied depending on the data type:

Numerical Columns

Missing values were replaced using the average value of the column.

Example:

=AVERAGEIF(A2:A601,"<>")

Categorical Columns

Blank values were replaced using the most frequent value (mode) in the column.

Example:

- Department blanks replaced with Sales (most frequent value).

4.3 Standardizing Data Formats

Inconsistent text values were standardized to maintain uniformity.

Examples:

- “male”, “Male”, “M” → standardized to Male
- “yes”, “YES”, “Yes” → standardized to Yes

Excel functions such as PROPER() and TRIM() were used to clean text data.

4.4 Filtering and Sorting

Filtering was used to analyze specific subsets of data such as employees who left the organization.

Sorting was applied to organize data based on attributes such as income or age.

4.5 Pivot Table Analysis

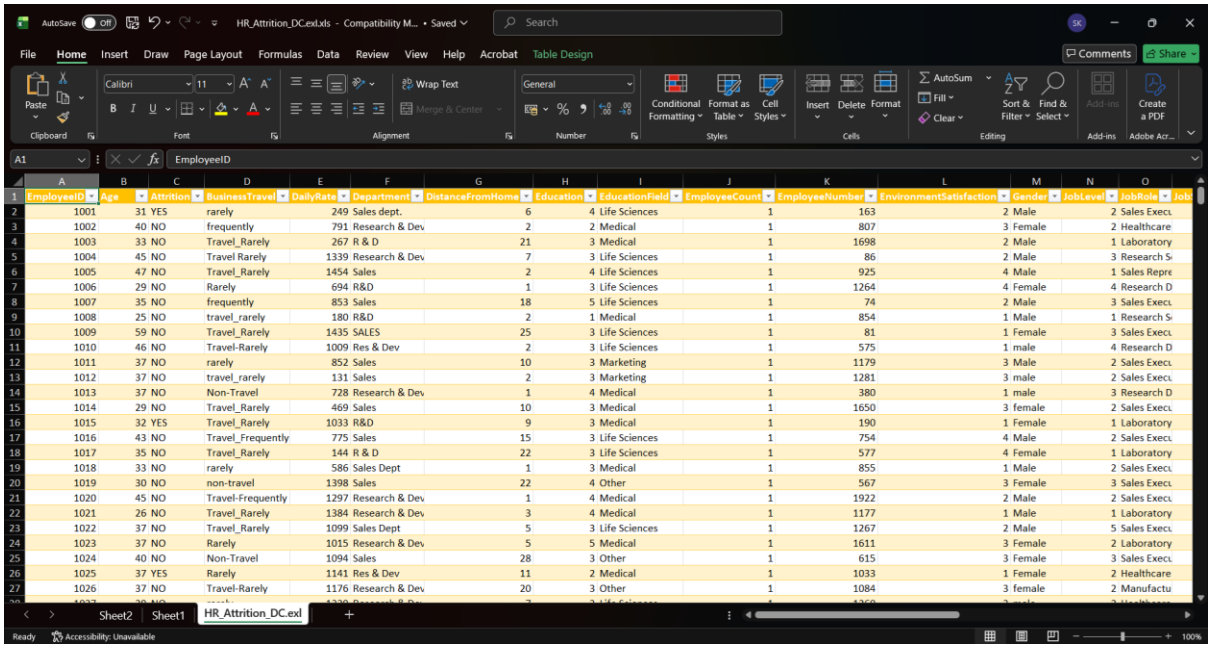
Pivot Tables were created to summarize and explore the dataset.

Example analyses:

- Employee count by department
- Average salary by job role
- Attrition distribution across departments

Pivot tables helped quickly identify trends in the dataset.

Dataset After Cleaning Excel :
https://docs.google.com/spreadsheets/d/13RX6NMriZUZWnUhDVAqzkNDYrStXGWEx/edit?usp=sharing&ouid=102166307444021995031&rtpof=true&sd=true



EmployeeID	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	Gender	JobLevel	JobRole
1001	31	YES	rarely	249	Sales dept.	6	4	Life Sciences	1	163	2	Male	2	Sales Execu
1002	40	NO	frequently	791	Research & Dev	2	2	Medical	1	807	3	Female	2	Healthcare
1003	33	NO	Travel_Rarely	267	R & D	21	3	Medical	1	1698	2	Male	1	Laboratory
1004	45	NO	Travel Rarely	1339	Research & Dev	7	3	Life Sciences	1	86	2	Male	3	Research S
1005	47	NO	Travel_Rarely	1454	Sales	2	4	Life Sciences	1	925	4	Male	1	Sales Repre
1006	29	NO	Rarely	694	R&D	1	3	Life Sciences	1	1264	4	Female	4	Research D
1007	35	NO	frequently	853	Sales	18	5	Life Sciences	1	74	2	Male	3	Sales Execu
1008	25	NO	travel_rarely	180	R&D	2	1	Medical	1	854	1	Male	1	Research S
1009	59	NO	Travel_Rarely	1435	SALES	25	3	Life Sciences	1	81	1	Female	3	Sales Execu
1010	46	NO	Travel-Rarely	1009	Res & Dev	2	3	Life Sciences	1	575	1	male	4	Research D
1011	37	NO	rarely	852	Sales	10	3	Marketing	1	1179	3	Male	2	Sales Execu
1012	37	NO	travel_rarely	131	Sales	2	3	Marketing	1	1281	3	male	2	Sales Execu
1013	37	NO	Non-Travel	728	Research & Dev	1	4	Medical	1	380	1	male	3	Research D
1014	29	NO	Travel_Rarely	469	Sales	10	3	Medical	1	1650	3	female	2	Sales Execu
1015	32	YES	Travel_Rarely	1033	R&D	9	3	Medical	1	190	1	Female	1	Laboratory
1016	43	NO	Travel_Frequently	775	Sales	15	3	Life Sciences	1	754	4	Male	2	Sales Execu
1017	35	NO	Travel_Rarely	144	R & D	22	3	Life Sciences	1	577	4	Female	1	Laboratory
1018	33	NO	rarely	586	Sales Dept	1	3	Medical	1	855	1	Male	2	Sales Execu
1019	30	NO	non-travel	1398	Sales	22	4	Other	1	567	3	Female	3	Sales Execu
1020	45	NO	Travel-Frequently	1297	Research & Dev	1	4	Medical	1	1922	2	Male	2	Sales Execu
1021	26	NO	Travel_Rarely	1384	Research & Dev	3	4	Medical	1	1177	1	Male	1	Laboratory
1022	37	NO	Travel_Rarely	1099	Sales Dept	5	3	Life Sciences	1	1267	2	Male	5	Sales Execu
1023	37	NO	Rarely	1015	Research & Dev	5	5	Medical	1	1611	3	Female	2	Laboratory
1024	40	NO	Non-Travel	1094	Sales	28	3	Other	1	615	3	Female	3	Sales Execu
1025	37	YES	Rarely	1141	Res & Dev	11	2	Medical	1	1033	1	Female	2	Healthcare
1026	37	NO	Travel-Rarely	1176	Research & Dev	20	3	Other	1	1084	3	female	2	Manufactu

---

5. Data Visualization using Power BI

After preprocessing the dataset, the cleaned data was imported into Power BI to build an interactive dashboard.

Power BI provides powerful visualization tools that help convert raw data into meaningful insights.

PowerBI : https://drive.google.com/file/d/1rb1mQebJqiGK_286trE0LTCNBhs-03U0/view?usp=drive_link

PowerBI PDF : https://drive.google.com/file/d/1w9gl4-W-fAHM7v5xDFgAN4SWaCkMwwm8/view?usp=drive_link

---

5.1 Dashboard Components

The dashboard contains multiple visualizations including:

Card Visuals

Used to display key metrics such as:

- Total Employees
- Employees who left
- Average salary

Bar Chart

Shows employee attrition by department.

This visualization helps identify which departments experience higher employee turnover.

Pie Chart

Displays the percentage distribution of attrition.

This chart provides a quick overview of how many employees stayed versus those who left.

Column Chart

Shows average salary by job role.

This helps analyze salary differences across roles.

Line Chart

Displays trends such as age distribution or attrition trends.

5.2 Interactive Features

To make the dashboard interactive, several features were used:

- Slicers for filtering by department, gender, and job role
- Filters to analyze attrition-specific records
- Drill-down functionality for deeper insights

These features allow users to explore the data dynamically.

---

6. Key Insights

The dashboard analysis revealed several insights:

- Certain departments show higher attrition rates.
- Employees working overtime are more likely to leave.
- Salary differences exist across job roles.
- Mid-level employees show higher turnover compared to others.

These insights help organizations understand employee retention patterns.

---

7. Conclusion

This mini project demonstrated how data preprocessing and visualization techniques can be used to extract insights from employee data.

Excel was used to clean and transform the dataset, while Power BI was used to create an interactive dashboard. The analysis highlighted factors influencing employee attrition and provided a clear visual representation of employee trends.

This project demonstrates the importance of data analytics in improving decision-making and understanding organizational patterns.